Curriculum Vitae

Okuto Morikawa

Personal Data

Date of Birth	1993.10.25
Place of Birth	Nagoya, Aichi, Japan
Nationality	Japan
Address	Center for Interdisciplinary Theoretical and Mathematical Sciences (iTHEMS), RIKEN, Wako 351-0198, Japan
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GitHub	`https://github.com/o-morikawa`
URLs	INSPIRE, ResearchMap, ResearchGate

Career History

2025.4-present	Special Postdoctoral Researcher, Center for Interdisciplinary Theoretical and Mathematical Sciences (iTHEMS), RIKEN
2025.4-present	Part-time Lecturer, Takushoku University
2025.2-3	Visiting Researcher, Department of Mathematical Sciences, Durham University
2024.4-2025.3	Special Postdoctoral Researcher, Interdisciplinary Theoretical and Mathematical Sciences Program (iTHEMS), RIKEN
2023.10-2024.3	Specially Appointed Researcher, Department of Physics, Osaka University
2021.4-2024.3	Postdoctoral Fellow of Japan Society for the Promotion of Science, Department of Physics, Osaka University
2018.4-2021.3	Research Fellow of Japan Society for the Promotion of Science, Department of Physics, Kyushu University

Academic Degrees

2021.3.24 **Ph.D. (Doctor of Science)**, Department of Physics, Kyushu University

Thesis Title:	"Numerical study of infrared criticality of the supersymmetric Landau–Ginzburg model"
Thesis Advisor:	Prof. Hiroshi Suzuki (Kyushu University)
DOI:	10.13140/RG.2.2.36216.42246
HDL:	2324/4474929

2018.3.20 **M.S.**, Department of Physics, Kyushu University

2016.3.25 **B.S.**, Department of Physics, Kyushu University

Awards

2026.9 Particle Physics Medal: Young Scientist Award in Theoretical Particle Physics, Japan Particle and Nuclear Theory Forum
O. Morikawa and S. Ogawa,
"Unified exact WKB framework for resonance — Zel'dovich/complex-scaling regularization and rigged Hilbert space,"
J. High Energy Phys. **10**, 49 (2025).

2026.2 Wolfram Community Staff Picks (Featured Contributor Badge), Wolfram Research, *Editorial selection / featured post*

2024.9 Seitaro Nakamura Prize, Soryushi Shogakukai
Research Title: "Non-perturbative foundation of generalized symmetries —Wilsonian lattice regularized point of view—"

2024.3 Young Scientist Award of the Physical Society of Japan, The Physical Society of Japan
Research Title: "Studies on generalized symmetries in lattice field theory"

2023.3 Prog. Theor. Exp. Phys. Editors' Choice Award, The Physical Society of Japan
M. Abe, O. Morikawa and H. Suzuki,
"Fractional topological charge in lattice Abelian gauge theory,"
Prog. Theor. Exp. Phys. **2023**, no.2, 023B03 (2023).
Picked up in
The Physical Society of Japan Hot Topics **3**, 010 (2023),
The Physical Society of Japan BUTSURI **78**, no.7, 419 (2023).

2021.9 Particle Physics Medal: Young Scientist Award in Theoretical Particle Physics, Japan Particle and Nuclear Theory Forum
O. Morikawa and H. Takaura,
"Identification of perturbative ambiguity canceled against bion,"
Phys. Lett. B **807**, 135570 (2020).

2021.3 Representative of the graduates (Ph.D.), Kyushu University
Valedictory speech (Video)

2021.3 11th Ikushi Prize, Japan Society for the Promotion of Science
Research Title: "Non-pertubative approaches to quantum field theory and superstring theory"

2016.3 Representative of the graduates (Bachelor of Science), Kyushu University

2012.3 Excellence Award of Shinkawa High School, Shinkawa High School, Aichi

Grants and Rewards

2026.4-2028.3	COREnet project of RCNP, The University of Osaka
2025.4-2030.3	JSPS Grant-in-Aid for Scientific Research (KAKENHI) Grant Number: JP25K17402
2025.4-2027.3	RIKEN FY2025 Incentive Research Projects
2024.4-2027.3	RIKEN Special Postdoctoral Researcher Program
2021.4-2024.3	JSPS Grant-in-Aid for Scientific Research (KAKENHI) Grant Number: JP21J30003 (2021-2022), JP22KJ2096 (2023)
2018.4-2021.3	JSPS Grant-in-Aid for Scientific Research (KAKENHI) Grant Number: JP18J20935
2016.4-2018.3	JASSO Scholarship "Exemption from Return for Particularly Outstanding Achievement"
2014.10-2015.3	Kyushu University Honors Scholarship
2014.4-2014.9	Kyushu University Honors Scholarship

Professional Memberships

2017-present	Japan Particle and Nuclear Theory Forum
2016-present	The Physical Society of Japan

Languages

Japanese (Native) and English

Conference/Symposium Organizer

2025.3.10-14	Organizer, chair and editor, Kyushu IAS-iTHEMS conference: Non-perturbative methods in QFT, Institute for Advanced Study, Kyushu University
2024.11-present	Organizer and chair, GRAPPA-QFT Seminar Series, `https://sites.google.com/view/grappa-qft`

Professional Service

2026-present	Eur. Phys. J. Plus (Springer), Referee
2026-present	MathSciNet (American Mathematical Society), Reviewer for Mathematical Reviews
2026-present	SciPost Physics Core (SciPost Foundation), Referee
2026-present	J. High Energy Phys. (Springer), Referee
2026-present	Next Research (Elsevier), Referee
2026-present	RIKEN Scientists' Assembly Steering Committee, Reviewer
2025-2026	Int. J. Mod. Phys. A (World Scientific), Editor

Teaching Experience

Lectures

2026.10-2027.3	"Science Laboratory," Undergraduate class in Takushoku University
2026.10-2027.3	"Physics II," Undergraduate class in Takushoku University
2026.4-9	"Science Laboratory," Undergraduate class in Takushoku University
2026.4-9	"Physics I," Undergraduate class in Takushoku University YouTube Playlist
2025.10-2026.3	"Science Laboratory," Undergraduate class in Takushoku University
2025.10-2026.3	"Physics II," Undergraduate class in Takushoku University YouTube Playlist
2025.4-9	"Science Laboratory," Undergraduate class in Takushoku University
2025.4-9	"Physics I," Undergraduate class in Takushoku University YouTube Playlist

Teaching Assistant

2017.4-9	"Exercise in Mechanics," Undergraduate class in Department of Physics, Kyushu University
2016.10-2017.3	"Exercise in Fundamental Physics," Undergraduate class in Department of Physics, Kyushu University
2016.4-9	"Exercise in Mechanics," Undergraduate class in Department of Physics, Kyushu University

Student Mentorship

2022.4-present	(Master & PhD) Soma Onoda, Kyushu University
2021.4-2026.3	(Master & PhD) Motokazu Abe [Dissertation; to appear on the Kyushu University website], Kyushu University
2023.4-2024.3	(Master) Yamato Honda, Kyushu University
2021.4-2024.3	(PhD) Yuta Nagoya [Dissertation] and (Master & PhD) Hiroki Wada [Dissertation], Osaka University
2021.4-2022.3	(Master) Mizuki Tanaka, Osaka University
2019.4-2021.3	(Master) Masahiro Ashie and Kengo Takeuchi, Kyushu University
2018.4-2020.3	(Master) Kosuke Ishikawa, Akira Nakayama and Kazuya Shibata, Kyushu University
2016.4-2018.3	(Undergraduate) Kiichi Goto, Hayato Kanno and Shintaro Toita, Kyushu University

Visits (more than 2 weeks)

2025.2-3	Durham University, UK
2024.10-11	YITP, Kyoto University, Japan
2024.8	NCTS, National Taiwan University, Taiwan
2020.9	YITP, Kyoto University, Japan
2019.1	Sogang University, Korea

Computer Skills

Efficient programming skills in C, C++, FORTRAN 77, Fortran 90/95, Python, Julia, Maxima, SageMath, Mathematica, Emacs Lisp.

Social Service

2026 Wolfram Community, Contributor
"Nonperturbative formulation of resonances in quantum mechanics based on exact WKB method"
`https://community.wolfram.com/groups/-/m/t/3635564`

2025 Mathematical Exchange Platform Bridging RIKEN and Kyoto University MACS, Lecturer

2022 Interview Survey of JSPS Ikushi Prize, Filming support, Consultant, and Official expert

Other Activities

2026.10	Review (Mathematical Reviews / MathSciNet) of J. A. Borrego-Morell and B. Shapiro, "Semiclassical expansion for exactly solvable differential operators," J. Phys. A: Math. Theor., 59 (2026), no. 29, 295202
2026.9	Review (Mathematical Reviews / MathSciNet) of N. Seiberg and W. Zhang, "Tori, Klein bottles, and modulo 8 parity/time-reversal anomalies of 2+1d staggered fermions," J. High Energy Phys., 2026, no. 5, 264
2026.8	Review (Mathematical Reviews / MathSciNet) of M. M. Anber, "Anomalies on ALE spaces and phases of gauge theory," J. High Energy Phys., 2026, no. 6, 149
2026.4.9	Referee Report on "A discrete formulation for three-dimensional winding number," SciPost Physics Core 9, 026 (2026), DOI: 10.21468/SciPost.Report.13242
2026.4-present	Organizer, iTHEMS Math-Phys Study Group Seminar Series, RIKEN
2025.9.16-19	Session chair, 2025 Annual (80th) Meeting of the Physical Society of Japan, Hiroshima University
2025.7.28-8.1	Session chair, Strings and Fields 2025, YITP
2025.4-present	Organizer, iTHEMS-phys Study Group Seminar Series, RIKEN
2024.9.16-19	Session chair, 2024 Annual (79th) Meeting of the Physical Society of Japan, Hokkaido University
2023.7.31-8.4	Session chair, The 40th International Symposium on Lattice Field Theory (Lattice2023), Fermilab, US
2022.10-2023.10	Organizer, Particle Physics Theory Group Seminar Series, Osaka University
2021.9.14-17	Session chair, 2021 Autumn Meeting of the Physical Society of Japan, Online
2021.4-10	Organizer, Particle Physics Theory Group Seminar Series, Osaka University
2020.8.26-28	Session chair, YONUPA Summer School 2020, Online
2017.4-2021.3	Organizer, Particle Physics Theory Group Seminar Series, Kyushu University

Publication List

1 Journal Articles (peer-reviewed)

[1] H. Watanabe, I. Kanamori, O. Morikawa, Y. Nagai, Y. Tanizaki and A. Tomiya,
"Numerical Hint for Dyon Condensation at $\theta = 2\pi$ via Wilson–'t Hooft Loops in $SU(2)$ Yang–Mills Theory,"
J. High Energy Phys. **10**, 068 (2026) [arXiv:2606.13428 [hep-lat]].

[2] S. Ogawa, T. Hirose and O. Morikawa,
"Complex scaling approach to quasinormal modes of Schwarzschild and Reissner–Nordström black holes,"
Prog. Theor. Exp. Phys. **2026**, no.9, 093E03 (2026) [arXiv:2604.20442 [hep-th]].

[3] O. Morikawa, S. Ogawa and S. Onoda,
"Geometric phase from encircling an exceptional point of a quantum resonance in the complex-scaling method,"
Phys. Rev. A **113**, 052225 (2026) [arXiv:2512.24528 [quant-ph]].

[4] O. Morikawa and S. Ogawa,
"On continuum and resonant spectra from exact WKB analysis,"
J. Phys. A: Math. Theor. **59**, 225306 (2026) [arXiv:2508.09211 [quant-ph]].

[5] S. Onoda, O. Fukushima, R. Hamazaki and O. Morikawa,
"Eigenstate Thermalization Hypothesis with Projective Representation,"
J. High Energy Phys. **06**, 138 (2026) [arXiv:2509.01931 [hep-th]].

[6] O. Morikawa and S. Ogawa,
"Exact WKB method for radial Schrödinger equation,"
J. Phys. A: Math. Theor. **59**, 145202 (2026) [arXiv:2510.11766 [quant-ph]].

[7] O. Morikawa and S. Ogawa,
"Unified exact WKB framework for resonance —Zel'dovich/complex-scaling regularization and rigged Hilbert space—,"
J. High Energy Phys. **10**, 49 (2025) [arXiv:2505.02301 [hep-th]].

[8] O. Morikawa and S. Ogawa,
"Nonperturbative Formulation of Resonances in Quantum Mechanics Based on Exact WKB Method,"
Prog. Theor. Exp. Phys. **2025**, no.10, 103B01 (2025) [arXiv:2503.18741 [hep-th]].

[9] O. Morikawa and H. Suzuki,
"Direct Monte Carlo computation of the 't Hooft partition function,"
Prog. Theor. Exp. Phys. **2025**, no.6, 063B04 (2025) [arXiv:2501.07042 [hep-lat]].

[10] M. Abe, O. Morikawa and H. Suzuki,
"Monte Carlo simulation of the $SU(2)/\mathbb{Z}_2$ Yang–Mills theory,"
Prog. Theor. Exp. Phys. **2025**, no.6, 063B03 (2025) [arXiv:2501.00286 [hep-lat]].

[11] O. Morikawa, S. Onoda and H. Suzuki,
"Yet another lattice formulation of 2D $U(1)$ chiral gauge theory via bosonization,"
Prog. Theor. Exp. Phys. **2024**, no.6, 063B01 (2024) [arXiv:2403.03420 [hep-lat]].

[12] Y. Honda, O. Morikawa, S. Onoda and H. Suzuki,
"Lattice realization of the axial $U(1)$ non-invertible symmetry,"
Prog. Theor. Exp. Phys. **2024**, no.4, 043B04 (2024) [arXiv:2401.01331 [hep-lat]].

[13] N. Kan, O. Morikawa, Y. Nagoya and Hiroki Wada
"Erratum to: Higher-group structure in lattice Abelian gauge theory under instanton-sum modification,"
Eur. Phys. J. C **84**, no.1, 22 (2024).

[14] M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
"Magnetic operators in 2D compact scalar field theories on the lattice,"
Prog. Theor. Exp. Phys. **2023**, no.7, 073B01 (2023) [arXiv:2304.14815 [hep-lat]].

[15] M. Abe, O. Morikawa and S. Onoda,
"Note on lattice description of generalized symmetries in $SU(N)/\mathbb{Z}_N$ gauge theories,"
Phys. Rev. D **108**, 014506 (2023) [arXiv:2304.11813 [hep-th]].

[16] M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
"Topology of $SU(N)$ lattice gauge theories coupled with $\mathbb{Z}_N$ 2-form gauge fields,"
J. High Energy Phys. **08**, 118 (2023) [arXiv:2303.10977 [hep-lat]].

[17] N. Kan, O. Morikawa, Y. Nagoya and H. Wada,
"Higher-group structure in lattice Abelian gauge theory under instanton-sum modification,"
Eur. Phys. J. C **83**, no.6, 481 (2023) [arXiv:2302.13466 [hep-th]].

[18] O. Morikawa, H. Wada and S. Yamaguchi,
"Phase structure of linear quiver gauge theories from anomaly matching,"
Phys. Rev. D **107**, 045020 (2023) [arXiv:2211.12079 [hep-th]].

[19] M. Abe, O. Morikawa and H. Suzuki,
"Fractional topological charge in lattice Abelian gauge theory,"
Prog. Theor. Exp. Phys. **2023**, no.2, 023B03 (2023) [arXiv:2210.12967 [hep-th]].

[20] H. Makino, O. Morikawa and H. Suzuki,
"Gradient flow and the Wilsonian renormalization group flow,"
Prog. Theor. Exp. Phys. **2021**, no.9, 099201 (2021).

[21] M. Ashie, O. Morikawa, H. Suzuki and H. Takaura,
"More on the infrared renormalon in $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$,"
Prog. Theor. Exp. Phys. **2020**, no.9, 093B02 (2020) [arXiv:2005.07407 [hep-th]].

[22] O. Morikawa and H. Takaura,
"Identification of perturbative ambiguity canceled against bion,"
Phys. Lett. B **807**, 135570 (2020) [arXiv:2003.04759 [hep-th]].

[23] K. Ishikawa, O. Morikawa, K. Shibata and H. Suzuki,
"Vacuum energy of the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$ in the $1/N$ expansion,"
Prog. Theor. Exp. Phys. **2020**, no.6, 063B02 (2020) [arXiv:2001.07302 [hep-th]].

[24] K. Ishikawa, O. Morikawa, A. Nakayama, K. Shibata, H. Suzuki and H. Takaura, "Infrared renormalon in the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$," Prog. Theor. Exp. Phys. **2020**, no.2, 023B10 (2020) [arXiv:1908.00373 [hep-th]].

[25] M. Ashie, O. Morikawa, H. Suzuki, H. Takaura and K. Takeuchi, "Infrared renormalon in $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$," Prog. Theor. Exp. Phys. **2020**, no.2, 023B01 (2020) [arXiv:1909.05489 [hep-th]].

[26] K. Ishikawa, O. Morikawa, K. Shibata, H. Suzuki and H. Takaura, "Renormalon structure in compactified spacetime," Prog. Theor. Exp. Phys. **2020**, no.1, 013B01 (2020) [arXiv:1909.09579 [hep-th]].

[27] O. Morikawa, "Continuum limit in numerical simulations of the $\mathcal{N} = 2$ Landau–Ginzburg model," Prog. Theor. Exp. Phys. **2019**, no.10, 103B03 (2019) [arXiv:1906.00653 [hep-lat]].

[28] O. Morikawa, "Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model with two superfields," J. High Energy Phys. **12**, 045 (2018) [arXiv:1810.02519 [hep-lat]].

[29] A. Kasai, O. Morikawa and H. Suzuki, "Gradient flow representation of the four-dimensional $\mathcal{N} = 2$ super Yang–Mills supercurrent," Prog. Theor. Exp. Phys. **2018**, no.11, 113B02 (2018) [arXiv:1808.07300 [hep-lat]].

[30] O. Morikawa and H. Suzuki, "Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model," Prog. Theor. Exp. Phys. **2018**, no.8, 083B05 (2018) [arXiv:1805.10735 [hep-lat]].

[31] O. Morikawa and H. Suzuki, "Axial $U(1)$ anomaly in a gravitational field via the gradient flow," Prog. Theor. Exp. Phys. **2018**, no.7, 073B02 (2018) [arXiv:1803.04132 [hep-th]].

[32] H. Makino, O. Morikawa and H. Suzuki, "Gradient flow and the Wilsonian renormalization group flow," Prog. Theor. Exp. Phys. **2018**, no.5, 053B02 (2018) [arXiv:1802.07897 [hep-th]].

[33] H. Makino, O. Morikawa and H. Suzuki, "One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator," Prog. Theor. Exp. Phys. **2017**, no.6, 063B08 (2017) [arXiv:1704.04862 [hep-lat]].

[34] H. Makino and O. Morikawa, "Lorentz symmetry violation in the fermion number anomaly with the chiral overlap operator," Prog. Theor. Exp. Phys. **2016**, no.12, 123B06 (2016) [arXiv:1609.08376 [hep-lat]].

2 Preprints

[1] O. Morikawa and S. Ogawa, "A physical connection criterion for reaction-coordinate reduction in quantum-field tunneling," https://doi.org/10.5281/zenodo.23050742.

[2] O. Morikawa,
"Center-twisted Gribov spectra and the finite-volume Gaussian response in the refined Gribov–Zwanziger framework,"
arXiv:2609.03525 [hep-th].

[3] O. Morikawa,
"The Changing Meaning of One-Loop Exactness in $\mathcal{N} = 2$ Supersymmetric Yang–Mills Theory,"
https://doi.org/10.5281/zenodo.23168337.

[4] O. Morikawa,
"Two Languages for the Same Resonance: From Nuclear Decay to Black-Hole Ringdown,"
arXiv:2608.29806 [gr-qc].

[5] O. Morikawa, S. Ogawa and T. Hirose,
"Riesz–Laurent representation of black-hole scattering and sourced response at exceptional points,"
arXiv:2608.14752 [gr-qc].

[6] S. Ogawa, O. Morikawa and T. Hirose,
"Quasinormal modes and continuum response of de Sitter black holes via complex scaling method,"
arXiv:2605.03277 [hep-th].

[7] O. Morikawa and H. Suzuki,
"Winding number on 3D lattice,"
arXiv:2412.03888 [hep-lat].

3 Editorials / Prefaces

[1] O. Morikawa and O. Fukushima,
Preface,
Int. J. Mod. Phys. A **41**, Issue 07, 2602001 (2026),
Special Issue: *Non-perturbative Methods in Quantum Field Theory.*
https://www.worldscientific.com/toc/ijmpa/41/07

4 Thesis

[1] O. Morikawa
"Numerical study of infrared criticality of the supersymmetric Landau–Ginzburg model,"
Ph.D. Thesis, Department of Physics, Kyushu University, 2021.
DOI: 10.13140/RG.2.2.36216.42246
HDL: 2324/4474929

5 Proceedings

5.1 Lattice conference (peer-reviewed)

[1] A. Tomiya, H. Watanabe, I. Kanamori, O. Morikawa, Y. Nagai and Y. Tanizaki,
"Gradient-flow actions, integerness, and gluonic–fermionic topology matching in $SU(3)$

Yang–Mills theory,"
Proc. Sci. **LATTICE2026**, 305 (2027).

[2] H. Watanabe, I. Kanamori, O. Morikawa, Y. Nagai, Y. Tanizaki and A. Tomiya,
"Dyon Condensation at $\theta = 2\pi$ from Wilson–'t Hooft Loops in Yang–Mills Theory,"
Proc. Sci. **LATTICE2026**, 217 (2027) [arXiv:2610.31281 [hep-lat]].

[3] O. Morikawa and H. Suzuki,
"Direct numerical simulation of 't Hooft partition function and (de)confining phase,"
Proc. Sci. **LATTICE2025**, 182 (2026) [arXiv:2601.20159 [hep-lat]].

[4] O. Morikawa, S. Onoda and H. Suzuki,
"Novel Lattice Formulation of 2D Chiral Gauge Theory via Bosonization,"
Proc. Sci. **LATTICE2024**, 363 (2025) [arXiv:2501.18949 [hep-lat]].

[5] M. Abe and O. Morikawa,
"Numerical simulation of fractional topological charge in $SU(N)$ gauge theory coupled with $\mathbb{Z}_N$ 2-form gauge fields,"
Proc. Sci. **LATTICE2024**, 380 (2025) [arXiv:2501.11438 [hep-lat]].

[6] O. Morikawa, M. Tanaka, M. Kitazawa and H. Suzuki,
"Lattice study of RG fixed point based on gradient flow in 3D $O(N)$ sigma model,"
Proc. Sci. **LATTICE2024**, 349 (2025) [arXiv:2410.19425 [hep-lat]].

[7] M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
"Lattice construction of mixed 't Hooft anomaly with higher-form symmetry,"
Proc. Sci. **LATTICE2023**, 361 (2024) [arXiv:2401.00495 [hep-lat]].

[8] M. Abe, N. Kan, O. Morikawa, Y. Nagoya, S. Onoda and H. Wada,
"Higher-group symmetry in lattice gauge theories with restricted topological sectors,"
Proc. Sci. **LATTICE2023**, 365 (2024) [arXiv:2310.01787 [hep-lat]].

[9] O. Morikawa,
"Numerical study of ADE-type $\mathcal{N} = 2$ Landau–Ginzburg models,"
Proc. Sci. **LATTICE2019**, 145 (2020) [arXiv:1908.03411 [hep-lat]].

[10] H. Makino, O. Morikawa and H. Suzuki,
"One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator,"
EPJ Web Conf. **175**, 11013 (2018) [arXiv:1710.00536 [hep-lat]].

5.2 The Physical Society of Japan

[1] S. Ogawa, O. Morikawa and T. Hirose,
"Quasinormal modes and continuum response of de Sitter black holes via complex scaling method,"
Meeting Abstracts of the Physical Society of Japan **81(2)**, ROMBUNNO.14pG721-4 (2026).

[2] S. Ogawa, T. Hirose and O. Morikawa,
"Complex scaling approach to quasinormal modes of Schwarzschild and Reissner-Nordstrom black holes,"
Meeting Abstracts of the Physical Society of Japan **81(2)**, ROMBUNNO.14pG721-3 (2026).

[3] O. Morikawa, S. Ogawa and S. Onoda,
"Geometric phase of exceptional point as quantum resonance in complex scaling method ,"
Meeting Abstracts of the Physical Society of Japan **81(1)**, ROMBUNNO.23pS1-10 (2026).

[4] O. Morikawa and S. Ogawa,
"Complex Analytic Structure and Resurgence in the Radial Schrödinger Equation,"
Meeting Abstracts of the Physical Society of Japan **81(1)**, ROMBUNNO.23pS1-9 (2026).

[5] O. Morikawa and S. Ogawa,
"Unified exact WKB framework for quantum mechanical resonance,"
Meeting Abstracts of the Physical Society of Japan **80(2)**, ROMBUNNO.19aEK201-10 (2025).

[6] T. Fukui, O. Morikawa and H. Suzuki,
"Gradient flow for winding numbers in real-space representation,"
Meeting Abstracts of the Physical Society of Japan **80(2)**, ROMBUNNO.16aSK310-10 (2025).

[7] O. Fukushima, R. Hamazaki, O. Morikawa and S. Onoda,
"Eigenstate Thermalization Hypothesis with Projective Representation,"
Meeting Abstracts of the Physical Society of Japan **80(2)**, ROMBUNNO.16aEK201-1 (2025).

[8] O. Morikawa,
"Lattice non-perturbative understanding of generalized symmetries and applications,"
Meeting Abstracts of the Physical Society of Japan **80(1)**, ROMBUNNO.18aS1-2 (2025) [J-Global: 202502271580568708].

[9] O. Morikawa and S. Ogawa,
"Non-perturbative formulation of quasi-stationary resonant state based on exact WKB method,"
Meeting Abstracts of the Physical Society of Japan **79(2)**, ROMBUNNO.18pWB101-16 (2024) [J-Global: 202502278248352483].

[10] O. Morikawa, S. Onoda and H. Suzuki,
"Yet another lattice formulation of 2D $U(1)$ chiral gauge theory via bosonization,"
Meeting Abstracts of the Physical Society of Japan **79(2)**, ROMBUNNO.16pWB102-6 (2024) [J-Global: 202502234377619864].

[11] O. Morikawa,
"Studies on generalized symmetries in lattice field theory,"
Meeting Abstracts of the Physical Society of Japan **79(1)**, ROMBUNNO.18aS1-8 (2024) [J-Global: 202402217971446919].

[12] M. Abe, N. Kan, O. Morikawa, Y. Nagoya, S. Onoda and H. Wada,
"Higher-group symmetry in lattice gauge theories with restriction of instanton numbers,"
Meeting Abstracts of the Physical Society of Japan **78(2)**, ROMBUNNO.16aRA23-9 (2023) [J-Global: 202302260250037895].

[13] M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
"Lattice formulation of magnetic operators in 2D compact scalar theories,"
Meeting Abstracts of the Physical Society of Japan **78(2)**, ROMBUNNO.16aRA23-8 (2023) [J-Global: 202302241386562671].

[14] M. Abe, O. Morikawa, S. Onoda, H. Suzuki and Y. Tanizaki,
"Fractional topological charge in 4d lattice $SU(N)$ gauge theory,"
Meeting Abstracts of the Physical Society of Japan **78(2)**, ROMBUNNO.16aRA23-7 (2023) [J-Global: 202302255016737097].

[15] O. Morikawa, H. Wada and S. Yamaguchi,
"Constraints on the phase structure of linear quiver gauge theories from anomalies for one-form symmetries,"
Meeting Abstracts of the Physical Society of Japan **77(2)**, ROMBUNNO.6pA132-12 (2022) [J-Global: 202302285992771528].

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6 Other Publications

[1] O. Morikawa, S. Ogawa, and T. Hirose,
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online textbook/monograph, publicly available from the authors' repositories, 2026.
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[2] O. Morikawa and S. Ogawa,
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7 Public Repositories

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"CSMQNM.jl,"
GitHub: skeleton for computing black-hole quasinormal-mode candidates with the complex scaling method (CSM), `https://github.com/o-morikawa/CSMQNM.jl`, 2026.

[2] O. Morikawa,
"AutoTortoise.jl,"
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[3] O. Morikawa,
"ErrorEst.jl,"
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[4] O. Morikawa,
"HigherCup,"
GitHub: Symbolic manipulation of Steenrod higher cups, `https://github.com/o-morikawa/HigherCup`, 2024

[5] O. Morikawa,
"Wind3D,"
GitHub: Discrete approximation to 3D winding number mapping $T^3 \to U(N)$, `https://github.com/o-morikawa/Wind3D`, 2024.

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(a) O. Morikawa,
"HigherBfields.jl (forked from akio-tomiya/Gaugefields.jl),"
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"Gaugefields.jl,"
GitHub: Julia package for utilities of quantum theory of (higher-form) general gauge fields, `https://github.com/o-morikawa/Gaugefields.jl`, 2024.

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"Resummation,"
GitHub: Mathematica codes for some resummation techniques based on perturbation theory, `https://github.com/o-morikawa/Resummation`, 2022.

[9] O. Morikawa,
"LandauGinzburg,"
GitHub: C++ and Julia codeset for simulation of $\mathcal{N} = 2$ Landau–Ginzburg model, `https://github.com/o-morikawa/LandauGinzburg`, 2021.

List of Presentations

1 Conference activities, talks, and seminars

[1] "Mathematica Applications in Theoretical Particle Physics," **Invited Speaker**, Wolfram Virtual Conference Japan 2026, Dec 14, 2026

[2] "Quantum Resonance as a Global Connection Problem," Ikushi Prize Laureates' Networking Meeting, Nov 30, 2026

[3] "Center-twisted Gribov spectra and confinement problem," **Invited Seminar** at Theoretical Particle Theory Group in Kindai University, Nov 27, 2026

[4] "Resonances, Spectral Theory, and Resurgent Structure," 30th Interdisciplinary Exchange Evening at RIKEN, Nov 20, 2026

[5] "Center-Twisted Gribov Spectra and the Question of Confinement," **Invited Seminar** at Theoretical Particle Theory Group in Kyushu University, Nov 13, 2026

[6] "Analytic structure of quantum resonances via exact WKB, scattering theory, and complex scaling," **Invited Seminar** at Nishida Group in Institute of Science Tokyo, Nov 9, 2026

[7] "Resonances, Spectral Theory, and Resurgence from Exact WKB Analysis," **Invited Speaker**, New Developments in Microlocal Analysis and Asymptotic Analysis, Oct 29, 2026

[8] "Gradient-Flow Actions, Integerness, and Gluonic–Fermionic Topology Matching in SU(3) Yang–Mills Theory," *talk by H. Watanabe*, 2026 Annual (81st) Meeting of the Physical Society of Japan, Sep 16, 2026

[9] "Quasinormal modes and continuum response of de Sitter black holes via complex scaling method," *talk by T. Hirose*, 2026 Annual (81st) Meeting of the Physical Society of Japan, Sep 14, 2026

[10] "Complex scaling approach to quasinormal modes of Schwarzschild and Reissner–Nordstrom black holes," 2026 Annual (81st) Meeting of the Physical Society of Japan, Sep 14, 2026

[11] "Complex scaling approach to black-hole quasinormal modes and continuum response," Strings and Fields 2026, Aug 18, 2026

[12] "Gradient-Flow Actions, Integerness, and Gluonic–Fermionic Topology Matching in SU(3) Yang–Mills Theory," *talk by A. Tomiya*, The 43rd International Symposium on Lattice Field Theory (Lattice2026), Jul 29, 2026

[13] "Dyon Condensation at $\theta = 2\pi$ from Wilson–'t Hooft Loops in Yang–Mills Theory," The 43rd International Symposium on Lattice Field Theory (Lattice2026), Jul 29, 2026

[14] "Towards a nonperturbative formulation of non-Hermitian quantum physics," **Invited Speaker**, Particle-Physics Endeavor Across Kyushu (P^2EAK), Jul 20, 2026

[15] "Lattice 2D $U(1)$ chiral gauge theory and magnetically charged vertex operators via bosonization," **Invited Speaker**, Frontiers of Lattice Fermions, Jul 2, 2026

[16] "Infrared Renormalons, Bion Ambiguities, and Resurgence in Compactified Gauge Theories," **Invited Seminar** at Takushoku University, Mar 17, 2026

[17] "'t Hooft Loop in SU(2) Yang-Mills Theory Revisited," *talk by H. Watanabe*, 2026 Spring Meeting of the Physical Society of Japan, Mar 26, 2026

[18] "Geometric phase of exceptional point as quantum resonance in complex scaling method ," 2026 Spring Meeting of the Physical Society of Japan, Mar 23, 2026

[19] "Complex Analytic Structure and Resurgence in the Radial Schrödinger Equation," 2026 Spring Meeting of the Physical Society of Japan, Mar 23, 2026

[20] "Towards a nonperturbative formulation of non-Hermitian quantum physics: resonance, complex scaling, and resurgence analysis," **Invited Seminar** at Takushoku University, Jan 21, 2026

[21] "Unified exact WKB framework for quantum resonance and scattering theory," KEK Theory Workshop 2025, Dec 16, 2025

[22] "Study of confinement phase based on lattice QCD simulation," Ikushi Prize Laureates' Networking Meeting, Nov 28, 2025

[23] "What is quantum resonance? —a viewpoint from resurgence theory—," **Invited Seminar** at Theoretical Particle Theory Group in Kobe University, Nov 26, 2025

[24] "Exact WKB framework for quantum resonance and spectral theory," **Invited Seminar** at Particle Theory and Cosmology Group in Tohoku University, Nov 21, 2025

[25] "Direct numerical simulation of 't Hooft partition function and (de)confining phase," The 42nd International Symposium on Lattice Field Theory (Lattice2025), Nov 4, 2025

[26] "Non-perturbative Confirmation of the Wilson–Fisher Fixed Point via Gradient Flow in 3D $O(N)$ Sigma Model," **Invited Seminar** at Theoretical Particle Physics Group in Kyoto University, Oct 29, 2025

[27] "Resonances, Spectral Theory, and the Resurgent Structure," **Invited Speaker**, Japan-UK Workshop on Quantum Gravity, Sep 23, 2025

[28] "Unified exact WKB framework for quantum mechanical resonance," 2025 Annual (80th) Meeting of the Physical Society of Japan, Sep 19, 2025

[29] "Lattice gradient flows (de-)stabilizing topological sectors for $SU(3)$," *talk by A. Tomiya*, 2025 Annual (80th) Meeting of the Physical Society of Japan, Sep 18, 2025

[30] "Hybrid Monte Carlo Method with Dynamical Flux in Lattice $SU(2)/\mathbb{Z}_2$ Theory and Resolution of Topological Freezing," *talk by M. Abe*, 2025 Annual (80th) Meeting of the Physical Society of Japan, Sep 17, 2025

[31] "Eigenstate Thermalization Hypothesis with Projective Representation," *talk by S. Onoda*, 2025 Annual (80th) Meeting of the Physical Society of Japan, Sep 16, 2025

[32] "Gradient flow for winding numbers in real-space representation," *talk by T. Fukui*, 2025 Annual (80th) Meeting of the Physical Society of Japan, Sep 16, 2025

[33] "Non-perturbative analysis of resonance physics by exact WKB method," Finite temperature field theory and its application 2025 (TQFT 2025), Sep 3, 2025

[34] "Non-perturbative understanding of resonant phenomena based on exact WKB analysis," Strings and Fields 2025, Aug 1, 2025

[35] "Lattice simulation of Wilson–'t Hooft classification in $SU(2)$ Yang–Mills theory with θ term," **Poster**, iTHEMS NOW & NEXT 2025, Jul 24, 2025

[36] "Exact WKB as unified analytic structure for resonance physics," **Invited Seminar** at Math-Phys Working Group in RIKEN iTHEMS, Jun 27, 2025

[37] "Non-perturbative formulation of resonant quantum mechanics within unified exact WKB framework," **Invited Seminar** at Asymptotics in Astrophysics Working Group in RIKEN iTHEMS, May 28, 2025

[38] "Winding number on 3D lattice," JuliaQCD User Briefing, May 16, 2025

[39] "Winding number on 3D lattice," **Invited Seminar** at CREST Research Seminar on "Theoretical studies of topological phases of matter," May 2, 2025

[40] "Hybrid Monte Carlo simulation of the $SU(2)/\mathbb{Z}_2$ Yang–Mills theory and resolution of topological freezing," *talk by M. Abe*, 2025 Spring Meeting of the Physical Society of Japan, Mar 18, 2025

[41] "Lattice non-perturbative understanding of generalized symmetries and applications," **Invited Speaker**, 2025 Spring Meeting of the Physical Society of Japan, Mar 18, 2025

[42] "Computational lattice foundation of generalized symmetry and its applications," **Invited Seminar** at Theoretical Particle Physics Group in University of Tsukuba, Nov 22, 2024

[43] "Lattice non-perturbative foundation of generalized symmetry and applications," **Invited Seminar** at Theoretical Particle Physics Group in Shinshu University, Nov 19, 2024

[44] "Non-perturbative formulation of quasi-stationary resonant state based on exact WKB method," 2024 Annual (79th) Meeting of the Physical Society of Japan, Sep 18, 2024

[45] "Yet another lattice formulation of 2D $U(1)$ chiral gauge theory via bosonization," *talk by S. Onoda*, 2024 Annual (79th) Meeting of the Physical Society of Japan, Sep 16, 2024

[46] "Applications of numerical computations of the fractional topological charge," *talk by M. Abe*, 2024 Annual (79th) Meeting of the Physical Society of Japan, Sep 16, 2024

[47] "Complete implementation of generalized symmetries on the lattice," **Invited Speaker**, Discrete Approaches to the Dynamics of Fields and Space–Time 2024, Sep 3, 2024

[48] "Lattice implementation of generalized symmetry and 't Hooft anomaly," **Invited Speaker**, NCTS-iTHEMS Joint Workshop on Matters to Spacetime: Symmetries and Geometry, Aug 29, 2024

[49] "Novel Lattice Formulation of 2D Chiral Gauge Theory via Bosonization," *talk by S. Onoda*, The 41st International Symposium on Lattice Field Theory (Lattice2024), Aug 2, 2024

[50] "Numerical simulation of fractional topological charge in $SU(N)$ gauge theory coupled with $\mathbb{Z}_N$ 2-form gauge fields," *talk by M. Abe*, The 41st International Symposium on Lattice Field Theory (Lattice2024), Aug 1, 2024

[51] "Lattice study of RG fixed point based on gradient flow in 3D $O(N)$ sigma model," The 41st International Symposium on Lattice Field Theory (Lattice2024), Jul 30, 2024

[52] "Symmetry and anomaly in lattice gauge theory," **Poster**, iTHEMS NOW & NEXT 2024, Jul 19, 2024

[53] "Towards computational foundations of generalized symmetries," **Invited Seminar** at Field Theory Research Team in RIKEN R-CCS, Apr 26, 2024

[54] "Generalized symmetries and topological phenomena in lattice gauge theory," **Invited Speaker**, Komaba Workshop on Non-perturbative Approaches to Quantum Field Theory, Mar 30, 2024

[55] "Lattice realization of the axial $U(1)$ non-invertible symmetry," *talk by Y. Honda*, 2024 Spring Meeting of the Physical Society of Japan, Mar 19, 2024

[56] "Studies on generalized symmetries in lattice field theory," **Invited Speaker**, 2024 Spring Meeting of the Physical Society of Japan, Mar 18, 2024

[57] "Lattice realization of the axial $U(1)$ non-invertible symmetry," **Invited Seminar** at Yukawa Institute for Theoretical Physics (YITP) in Kyoto University, Feb 16, 2024

[58] "Lattice study of RG fixed point based on gradient flow in 3D $O(N)$ sigma model," KEK Theory Workshop 2023, Nov 30, 2023

[59] "Topological charge fractionality and higher-group of lattice higher-form gauge fields," **Invited Seminar** at Hadron Theory Group in the University of Tokyo, Nov 9, 2023

[60] "Lattice description of generalized symmetries and topological phenomena," **Invited Seminar** at KEK Theory Center, Oct 3, 2023

[61] "How to construct generalized symmetries and 't Hooft anomaly from lattice viewpoint," The International Workshop on Theoretical Particle Physics 2023 (HET Camp 2023), Sep 26, 2023

[62] "Higher-group symmetry in lattice gauge theories with restriction of instanton numbers," 2023 Annual (78th) Meeting of the Physical Society of Japan, Sep 16, 2023

[63] "Lattice formulation of magnetic operators in 2D compact scalar theories," *talk by S. Onoda*, 2023 Annual (78th) Meeting of the Physical Society of Japan, Sep 16, 2023

[64] "Fractional topological charge in 4d lattice $SU(N)$ gauge theory," *talk by M. Abe*, 2023 Annual (78th) Meeting of the Physical Society of Japan, Sep 16, 2023

[65] "Higher-group symmetry in lattice gauge theories with restricted topological sectors," The 40th International Symposium on Lattice Field Theory (Lattice2023), Jul 31, 2023

[66] "Lattice construction of mixed 't Hooft anomaly with higher-form symmetry," *talk by M. Abe*, The 40th International Symposium on Lattice Field Theory (Lattice2023), Jul 31, 2023

[67] "Formulation of fractional topological charge in lattice gauge theory," *talk by M. Abe*, 2022 Autumn Meeting of the Physical Society of Japan, Sep 8, 2022

[68] "Constraints on the phase structure of linear quiver gauge theories from anomalies for one-form symmetries," *talk by H. Wada*, 2022 Autumn Meeting of the Physical Society of Japan, Sep 6, 2022

[69] "Analysis of Fixed Point in 3-Dimensional $O(N)$ Scalar Model Using Gradient Flow," *talk by M. Tanaka*, 2022 Annual (77th) Meeting of the Physical Society of Japan, Mar 15, 2022

[70] "Numerical study of deformation of target space for the $\mathcal{N} = 2$ Landau–Ginzburg models," 2021 Autumn Meeting of the Physical Society of Japan, Sep 14, 2021

[71] "Counterargument to the expected resurgence structure concerning bion and new understanding," *talk by H. Takaura*, 2021 Annual (76th) Meeting of the Physical Society of Japan, Mar 12, 2021

[72] "Perturbative ambiguities and resurgence in circle-compactified theories," **Invited Seminar** at Theoretical Particle Physics Group in Tokyo Institute of Technology, Jan 20, 2021

[73] "Perturbative ambiguities in compactified spacetime and resurgence structure," **Invited Speaker**, Potential Toolkit to Attack Nonperturbative Aspects of QFT -Resurgence and related topics-, Sep 24, 2020

[74] "Renormalon in $SU(N)$ QCD(adj.) on compactified spacetime," 2020 Autumn Meeting of the Physical Society of Japan, Sep 16, 2020

[75] "Identification of perturbative ambiguity canceled against bion," 2020 Autumn Meeting of the Physical Society of Japan, Sep 14, 2020

[76] "Resurgence structure in S^1-compactified spacetime," YONUPA Summer School 2020, Aug 28, 2020

[77] "Perturbative ambiguity in compactified spacetime and resurgence structure," **Poster**, Finite temperature field theory and its application 2020 (TQFT 2020), Aug 25, 2020

[78] "Resurgence structure on compactified spacetime with twisted boundary condition," Asia-Pacific Symposium for Lattice Field Theory (APLAT 2020), Aug 4, 2020

[79] "Computer simulation of the $\mathcal{N} = 2$ Landau–Ginzburg model," **Invited Seminar** at Theoretical Particle Physics Group in Osaka University, Jun 9, 2020

[80] "IR renormalon in the $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$," 2020 Annual (75th) Meeting of the Physical Society of Japan, Mar 16, 2020

[81] "IR renormalon in $SU(N)$ QCD(adj.) on $\mathbb{R}^3 \times S^1$," *talk by H. Takaura*, 2020 Annual (75th) Meeting of the Physical Society of Japan, Mar 16, 2020

[82] "Vacuum energy of the SUSY $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$," **Invited Speaker**, CPˆN model: recent developments and future directions, Jan 23, 2020

[83] "Infrared renormalon in the $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$," RIKEN-Kyushu workshop on particle, nuclear and astrophysics, Dec 23, 2019

[84] "Infrared renormalon in the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$," **Poster**, KEK Theory Workshop 2019, Dec 5, 2019

[85] "Continuum extrapolation and precision measurement of the scaling dimension in the $\mathcal{N} = 2$ Landau–Ginzburg model," 2019 Autumn Meeting of the Physical Society of Japan, Sep 18, 2019

[86] "Numerical study of ADE-type $\mathcal{N} = 2$ Landau–Ginzburg models," The 37th International Symposium on Lattice Field Theory (Lattice2019), Jun 23, 2019

[87] "Gradient flow representation of the 4D $\mathcal{N} = 2$ SYM supercurrent," 2019 Annual (74th) Meeting of the Physical Society of Japan, Mar 17, 2019

[88] "Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model," **Poster**, The 4th International Workshop on "Higgs as a Probe of New Physics" (HPNP2019), Feb 21, 2019

[89] "Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model," **Poster**, KEK Theory Workshop 2018, Dec 19, 2018

[90] "Numerical study of the $\mathcal{N} = 2$ Landau–Ginzburg model," iTHEMS-Kyushu WS -from particles and nuclei to cosmos-, Nov 21, 2018

[91] "Numerical study of $\mathcal{N} = 2$ Landau–Ginzburg models with multiple superfields," 2018 Autumn Meeting of the Physical Society of Japan, Sep 15, 2018

[92] "Gradient flow and the Wilsonian renormalization group flow," **Invited Seminar** at KEK Theory Center, May 20, 2018

[93] "Numerical study of $\mathcal{N} = 2$ Landau–Ginzburg models," 2018 Annual (73rd) Meeting of the Physical Society of Japan, Mar 23, 2018

[94] "Gravitational axial $U(1)$ anomaly via the gradient flow," *talk by H. Suzuki*, 2018 Annual (73rd) Meeting of the Physical Society of Japan, Mar 23, 2018

[95] "4D $\mathcal{N} = 2$ SUSY super current in terms of the gradient flow," *talk by A. Kasai*, 2018 Annual (73rd) Meeting of the Physical Society of Japan, Mar 23, 2018

[96] "Numerical study of the two-dimensional supersymmetric Landau–Ginzburg model," 123rd Meeting of the Kyushu Branch of the Physical Society of Japan, Dec 9, 2017

[97] "Conformal field theory and the Landau–Ginzburg description," **Invited Seminar** at Theoretical Particle Physics Group in Hokkaido University, Nov 17, 2017

[98] "Numerical study of $\mathcal{N} = 2$ Landau–Ginzburg models," **Invited Seminar** at Theoretical Particle Physics Group in Hokkaido University, Nov 17, 2017

[99] "Numerical simulation of the $\mathcal{N} = 2$ Landau–Ginzburg model," Setouchi Summer Institute 2017, Sep 28, 2017

[100] "Numerical simulation of $\mathcal{N} = 2$ Landau–Ginzburg model," 2017 Autumn Meeting of the Physical Society of Japan, Sep 14, 2017

[101] "Gradient flow and the Wilsonian renormalization group flow," *talk by H. Suzuki*, 2017 Autumn Meeting of the Physical Society of Japan, Sep 14, 2017

[102] "One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator," *talk by H. Suzuki*, 2017 Autumn Meeting of the Physical Society of Japan, Sep 14, 2017

[103] "One-loop perturbative coupling of A and $A_\star$ through the chiral overlap operator," The 35th International Symposium on Lattice Field Theory (Lattice2017), Jun 23, 2017

[104] "Lorentz symmetry violation in the fermion number anomaly with the chiral overlap operator," 2017 Annual (72nd) Meeting of the Physical Society of Japan, Mar 17, 2017

[105] "Lorentz symmetry violation in the fermion number anomaly with the chiral overlap operator," 122nd Meeting of the Kyushu Branch of the Physical Society of Japan, Dec 10, 2016

[106] "Lorentz symmetry violation in the fermion number anomaly with the chiral overlap operator," **Poster**, KEK Theory Workshop 2016, Dec 7, 2016

2 Other talks

[1] Coffee Meeting Talk at iTHEMS, Dec 12, 2025:
"Aspects of Quantum Resonance Revisited" (YouTube)

[2] Journal Club at Durham Univ., Feb 17, 2025:
"Lattice implementation of gauge field topology and generalized symmetry"

[3] Coffee Meeting Talk at iTHEMS, Sep 13, 2024:
"Gauge field topology on the lattice"

[4] Journal Club, Jan 18, 2024:

- Y. Honda, O. Morikawa, S. Onoda and H. Suzuki, "Lattice realization of the axial $U(1)$ non-invertible symmetry," Prog. Theor. Exp. Phys. **2024**, no.4, 043B04 (2024)

[5] Local Seminar at Osaka Univ., Apr 20, 2021:
"Perturbative ambiguities and resurgence in compactified spacetime"

[6] Journal Club, Jul 14, 2020:

- A. Carosso, A. Hasenfratz, and T. Neil, "Nonperturbative Renormalization of Operators in Near-Conformal Systems Using Gradient Flow," Phys. Rev. Lett. **121**, no.20, 201601 (2018)
- A. Carosso, "Stochastic Renormalization Group and Gradient Flow," J. High Energy Phys. **01**, 172 (2020)

[7] Journal Club, Oct 18, 2019:

- D. J. Gross and Y. Kitazawa, "A Quenched Momentum Prescription for Large N Theories," Nucl. Phys. B **206**, 440-472 (1982)

[8] Journal Club, Feb 25, 2019:

- A. Alexandru, G. Basar, P. F. Bedaque, G. W. Ridgway, and N. C. Warrington, "Sign problem and Monte Carlo calculations beyond Lefschetz thimbles," J. High Energy Phys. **05**, 053 (2016)

[9] Journal Club, May 14, 2018:

- M. Lüscher, P. Weisz, and U. Wolff, "A Numerical method to compute the running coupling in asymptotically free theories," Nucl. Phys. B **359**, 221-243 (1991)

[10] Journal Club, Sep 21, 2017:

- J. Polchinski, "String theory. Vol. 2: Superstring theory and beyond," Cambridge Monographs on Mathematical Physics, Cambridge University Press, 1998 [DOI: 10.1017/CBO9780511618123]
- T. Eguchi and Y. Sugawara, "Conformal field theory," Iwanami Shoten, Publishers, 2015

[11] Journal Club, Nov 4, 2016:

- D. M. Grabowska and D. B. Kaplan, "A Chiral Solution to the Ginsparg-Wilson Equation," Phys. Rev. D **94**, no.11, 114504 (2016)

Research Statement

My research spans from the non-perturbative formulation of quantum field theory (QFT) to the exploration of superstring theory, aiming to bridge the gap between lattice field theory and the elusive quest for a theory of everything. As quite important topics related to the theoretical structure of QFT itself, we have studied its non-perturbative formulation and quantum theory of gravity. For the non-perturbative dynamics in strongly coupled theories, a calculational method is provided by lattice field theory, which is the most well-developed framework or regularization so that spacetime is discretized as the set of points called the lattice. On the other hand, a candidate for a theory of everything unifying all fundamental forces including gravity may be provided by superstring theory. In superstring theory, we also expect to observe nontrivial and non-perturbative phenomena due to the string as a 1-dimensional object. However, there has been no definite non-perturbative formulation of superstring theory. In my research, I want to apply non-perturbative methods developed in lattice field theory to such nontrivial issues. Much many phenomena in QFT and superstring theory are subject to my interests. I hope that there are many potential benefits and insights that people have made no attempts to study this gap between lattice field theory and superstring theory.

I carried out the following four topics as explicit analyses of non-perturbative effects in QFT and superstring theory.

1 Numerical simulation of supersymmetric Landau–Ginzburg model

In the case that the target space of superstring theory is the Calabi–Yau manifold, we have a 2-dimensional $\mathcal{N} = 2$ superconformal field theory on the world sheet. The Landau–Ginzburg description of conformal field theory, that is, an infrared fixed point in a 2-dimensional supersymmetric theory (Landau–Ginzburg model) is believed to become superconformal. In this research, I want to achieve a triumph of lattice field theory, and pave the way for the numerical investigation of the Landau–Ginzburg model. In particular, as an example of Landau–Ginzburg model, the authors in Refs. [1, 2] simulated numerically the 2-dimensional $\mathcal{N} = 2$ massless Wess–Zumino model with a cubic superpotential, and then observed the infrared critical behavior.

In previous research, I established non-perturbative techniques for the Landau–Ginzburg model and verified, through numerical simulations, that its infrared fixed point corresponds to a superconformal field theory. Using the Kadoh–Suzuki formulation [3], which rigorously preserves supersymmetry, I applied numerical methods to various Landau–Ginzburg models belonging to the ADE classification and confirmed that the low-energy correlation functions correctly reproduce the expectations from the correspondence. Additionally, employing a finite-size scaling method considering the continuum limit, I performed precise measurements of correlation functions for the cubic potential, and provided an evidence for the restoration of locality.

As an example of simple target space, I considered the case of a torus given by $x^3+y^3+z^3$ and carried out numerical simulations under various deformations of the torus. Particularly, we can numerically verify the expectation that the central charge remains constant under the insertion of terms believed to represent a marginal operator xyz.

2 Studies on non-perturbative techniques based on gradient flow method

In lattice field theory, because of discretizing spacetime into a lattice, it has been a long posed problem to construct the energy-momentum tensor, a Noether current of spacetime symmetries. Recently, a method based on a kind of diffusion equation called gradient flow [4] has extensively developed. By deforming field configurations according to a diffusion equation in gradient flow, composite operators in terms of the flowed fields automatically become finite (renormalized). Similar problems arise for other symmetries such as supersymmetry that is incompatible with lattice spacetime structure.

Regarding this, the following research was studied:

- Although the Standard Model is a chiral gauge theory, we have not known any lattice formulation of chiral gauge theory except for the $U(1)$ gauge theory. A new lattice formulation using gradient flow was proposed by Grabowska and Kaplan [5]. I showed that the proposed lattice formulation faces difficulties to possess desired phenomena like the fermion number anomaly.

- I discussed and exemplified the relationship between gradient flow and Wilsonian renormalization group. Particularly, I analyzed the Banks–Zaks fixed point for N_f-flavor QCD approximated at the two-loop level; the Wilson–Fisher fixed point for the large N limit of the 3-dimensional $O(N)$ linear sigma model.

- I computed the chiral anomaly in a gravitational field from the chiral current and energy-momentum tensor constructed via gradient flow.

- Furthermore, using gradient flow, I constructed a universal expression for the supercurrent, a Noether current of supersymmetry, for the 4-dimensional $\mathcal{N} = 2$ supersymmetric Yang–Mills theory.

Currently, I am carrying out lattice simulations for the case of the 3-dimensional $O(N)$ sigma model and depicting a renormalization group flow diagram by using the gradient flow. Therefore I numerically and non-perturbatively demonstrate the existence of the Wilson–Fisher fixed point [6].

3 Studies on non-perturbative effects based on resurgence theory

Infrared renormalons, bion ambituities, and resurgent structure

In general, perturbative expansions in field theory lead to divergent series, introducing uncertainties in theoretical predictions. The proliferation of Feynman diagrams and contributions from instantons result in uncertainties in both perturbative and non-perturbative realms, which are believed to cancel each other out in a phenomenon known as resurgence structure. We also have known the existence of renormalon uncertainties in the perturbation theory. Recent attention has focused on the conjecture that the resurgence structure compensates for the renormalon uncertainty due to the presence of bions, semi-classical objects, in spacetime compactified on S^1 [7]. Despite vigorous discussions on this conjecture, several negative results have been reported, leading to a somewhat chaotic situation.

Focusing on the uncertainty in perturbation theory in spacetime compactified on S^1, I studied the following issues: (1) Using large N approximation, I analyzed the renormalon analysis in a spacetime where one direction is compactified, and showed that the renormalon uncertainty does not match that induced by bions. (2) I calculated the vacuum energy of the supersymmetric $\mathbb{C}P^{N-1}$ model on $\mathbb{R} \times S^1$ up to next-to-leading order in $1/N$ expansion,

and then demonstrated that renormalons itself cannot be discussed from this result. Especially this is quite different from the calculation of vacuum energy based on bions [8]. (3) I discussed how the infrared structure of perturbation theory is altered under S^1 compactification, by revealing that the explosion in the number of Feynman diagrams possesses an uncertainty enhanced by N, corresponding to bion uncertainty. We specified the resurgence structure. This mechanism naturally explains the inconsistency observed in the vacuum energy calculations. (4) I showed that, for $SU(N)$ gauge theories with adjoint fermions, renormalons do not exist for finite N.

Unified exact WKB framework of quantum resonance and scattering theory

My recent research is centered on a unified non-perturbative formulation of quantum resonance and quantum scattering theory based on exact WKB analysis and resurgence. Resonant states are ubiquitous in quantum physics, ranging from atomic and nuclear systems to open and non-Hermitian quantum systems, yet their theoretical description has long depended on a collection of distinct techniques such as phenomenological regularizations, complex scaling, and bound-state approximations. My goal is to place these approaches within a single analytic framework and thereby provide a mathematically controlled and physically transparent understanding of unstable quantum states with complex energies and finite decay widths. In this program, I extend exact WKB methods from ordinary bound-state problems to quasi-stationary and scattering phenomena, emphasizing the role of analyticity, spectral structure, and non-perturbative consistency.

In particular, I have studied how Zel'dovich regularization, the complex scaling method, and the rigged Hilbert space are interconnected when viewed through the exact WKB formalism. This perspective makes it possible to analyze resonance quantization conditions, Stokes structures, and resummed asymptotic solutions in a unified way, and it helps clarify why different resonance prescriptions successfully capture the same physical content. Through this line of research, I aim to connect exact WKB analysis with broader themes in spectral theory, non-Hermitian quantum mechanics, and resurgence. My long-term objective is to establish a general framework in which resonance and scattering phenomena can be treated non-perturbatively with both mathematical precision and practical applicability, and to extend these ideas toward more general problems in quantum theory and quantum field theory.

4 Lattice non-perturbative construction of generalized symmetries

Analysis of phase structures using generalized symmetry and the 't Hooft anomaly matching condition

In recent years, the concept of symmetry has been generalized, where the conventional 0-form symmetry associated with Noether's theorem is regarded as topological objects on a spacetime with a fixed time slice (in d dimensions). Furthermore, an extension of this symmetry allows for the discussion of topological objects extending $(d-p)$ dimensions for p-form symmetries [9, 10]. When such symmetries are gauged, leading to what is known as 't Hooft anomalies that quantum anomalies invariant under renormalization group flow, we can have nontrivial constraints on the phase structure in the strong coupling regime. This is expected to contribute to understanding the phase structure of QCD, and also many aspects not only of particle physics but also of condensed matter physics.

I analyzed the phase structure of linear quiver gauge theories based on the 't Hooft anomaly matching condition. In linear quiver gauge theories, $(K+1)$ $SU(N)$ gauge fields

are coupled with K Weyl fermions such that a Weyl fermion ψ_i is in the fundamental representation of the ith gauge group and the anti-fundamental representation of the $(i+1)$th gauge group (where $i = 1, 2, \ldots, K$). Particularly "direction" i constitutes a new dimension (dimensional deconstruction) [11]. It is interesting that this theory is quite similar to topological materials. I discussed scenarios that could occur in the strong coupling regime for K odd/even. A remarkable aspect of this analysis is that the gauge-invariant operator $\psi_1\psi_2\ldots\psi_K$ exhibits different types of condensation because it is either bosonic or fermionic if K is even or odd, respectively.

Research on generalized symmetry and topology in lattice regularized gauge theory

Most of the recent developments in research on generalized symmetry, such as those mentioned above, have remained largely in the realm of continuous spacetime formalism. It have not yet reached fully regularized discussions. Intricate techniques, viewpoints, and mathematics are employed to discuss non-perturbative effects. Despite the common use of lattice description when introducing generalized symmetries, the difficulties arises from dealing with "topology" on the lattice.

Actually, topology, an essential property of fiber bundle, has been constructed in lattice $SU(N)$ gauge theory in the 1980s [12]. An attempt to approach generalized symmetry from the lattice regularized perspective is quite nontrivial due to the complexity of this construction.

To address this, based on a simplified construction for the $U(1)$ gauge theory [13], I initially demonstrated the fractionality of topological charges and mixed 't Hooft anomalies by coupling $\mathbb{Z}_q$ 2-form gauge fields to the theory. It is however essential to note that introducing $\mathbb{Z}_q$ 2-form gauge fields is crucial for discussions on magnetic monopoles, but constructing monopoles is not possible on a lattice gauge configuration where topology is defined. We should assume the Witten effect happens. I have been developing methods to observe similar phenomena to the Witten effect on the lattice in 2-dimensional scalar field theories. I have applied it to bosonization of 2-dimensional chiral gauge theory, this approach awaiting further progress.

Moreover, under the $\mathbb{Z}_N$ 1-form gauge invariant construction which is the gauged center symmetry in lattice $SU(N)$ gauge theory, I proved the fractionality and the presence of mixed 't Hooft anomaly in non-Abelian gauge theories. This proof is completely realized in an ultra-local way. Particularly, while continuous theories are described by $\mathbb{Z}_N$ higher-form gauge fields as multiple $U(1)$ gauge fields and their breaking via Higgs mechanism, lattice theories are described by them simply as integer lattice fields; it provides a clearer understanding of such structures even in lattice gauge theory. We can achieve the realization of higher-group and non-invertible (categorical) symmetries in a more straightforward manner than in continuous theories.

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